
School of Computing, Engineering and Digital Technologies
Department of Data Science
Teesside University
Middlesbrough TS1 3BA

Generalisable Deep Learning for Cardiac Structure Segmentation Using Echocardiography

Analysis, Design and Implementation Report

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Student Name: Bharath Kumar Kapu

Supervisor: Abbas Haider

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ABSTRACT

This study examines the creation of generalisable deep learning models for the automatic segmentation of heart structures from echocardiographic images. Precise segmentation of essential cardiac structures, especially the left ventricle, is vital for the reliable computation of clinically significant metrics, such as end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF). However, manual segmentation is labour-intensive and vulnerable to inter-observer variability, underscoring the necessity for reliable automated alternatives.

The research uses the CAMUS dataset and assesses various deep learning architectures, such as U-Net, Attention U-Net, and U-Net++. Various training procedures were investigated, including diverse loss functions (cross-entropy, Dice loss, and focal loss) and data sampling methods to address class imbalance. This work primarily focuses on improving model generalisation by using strict patient-level data partitioning to prevent data leakage between training and test sets.

Experimental results indicate that the suggested methodology achieves strong segmentation performance, with Dice similarity coefficients approaching 0.90 and improved border identification, as measured by the 95th percentile Hausdorff Distance (HD95). The work extracts clinically relevant metrics (EDV, ESV, and EF) directly from the anticipated segmentation masks, in addition to segmentation.

The results highlight the importance of including appropriate loss functions, efficient sampling techniques, and architectural improvements to create durable and generalisable models. This study advances automated cardiac image analysis and provides a robust groundwork for the creation of clinically relevant deep learning systems in echocardiography.

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CHAPTER 1: INTRODUCTION

Cardiovascular diseases (CVDs) remain a primary cause of mortality globally, representing a substantial fraction of annual deaths worldwide. Timely diagnosis and precise monitoring of heart function are crucial for efficient clinical management and enhanced patient outcomes. Echocardiography has become one of the most common diagnostic methods due to its non-invasive characteristics, cost-effectiveness, and capacity for real-time visualisation of heart structures (Leclerc et al., 2019).

Echocardiographic imaging allows clinicians to evaluate the structural and functional characteristics of the heart. The assessment of the left ventricle (LV) is essential, as it is crucial in circulating oxygenated blood throughout the body. Clinical measures, including end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF), are often obtained from left ventricular analysis and are crucial to the identification of illnesses such as heart failure, cardiomyopathy, and other cardiovascular problems (Leclerc et al., 2019).

Despite its therapeutic significance, the extraction of these data is mainly dependent on manual segmentation, according to which doctors draw the perimeters of the cardiac structures in each frame of the echocardiographic sequence. This technique is fundamentally labour-intensive and requires a substantial degree of knowledge. This limitation has been consistently recognised in echocardiographic segmentation studies, where both expert variability and poor image quality render uniform contouring impossible (Leclerc et al., 2019; Kim et al., 2021). Furthermore, it is susceptible to inter-observer and intra-observer variability, potentially resulting in discrepancies in diagnosis and treatment planning. These constraints highlight the necessity for automated segmentation techniques that yield precise, dependable, and consistent outcomes.

In recent years, deep learning has transformed medical image analysis, enabling automated feature extraction and improving diagnostic workflows in healthcare (Esteva et al., 2019). Convolutional neural networks (CNNs) have exhibited exceptional efficiency in image segmentation tasks. Architectures like U-Net and its variants are extensively utilised for their capacity to capture both local and global contextual information, rendering them especially appropriate for medical imaging applications (Ronneberger et al., 2015). These developments contributed to the creation of automated devices proficient in accurately segmenting cardiac structures.

1.1 Problem Statement

Despite the promising outcomes of deep learning algorithms in cardiac image segmentation, several challenges exist. A principal challenge is in the quality and unpredictability of echocardiographic images. Echocardiography, in contrast to other imaging modalities, is significantly influenced by the operator's proficiency and patient-specific variables, leading to images characterised by differing degrees of noise, contrast, and artefacts. These variables limit segmentation methods from reliably identifying the precise borders of cardiac components.

A notable challenge is class disparity. In segmentation tasks, the background region generally constitutes a significantly larger proportion of the image than the region of focus, such as the left ventricle. This disparity may bias the learning process, leading models to prioritise background predictions and reducing their capacity to effectively classify smaller structures. Resolving this issue necessitates the use of specialised loss functions and sampling procedures that can balance the contribution of various classes during training.

Moreover, the problem of data leakage becomes a significant challenge in the creation of dependable models. Data leakage happens when information from the validation or test set is unintentionally utilised during training, resulting in too positive performance evaluations. In medical imaging, this frequently occurs when images from the same patient are incorporated into both training and validation datasets. Such techniques undermine the integrity of the review process and restrict the model's capacity to generalise to newer data. Consequently, the implementation of an appropriate patient-level data partitioning approach is crucial for guaranteeing an accurate performance evaluation.

The literature indicates that results obtained from different preprocessing pipelines, data partitions, and evaluation methodologies are challenging to compare equally, hence requiring a controlled and reproducible framework for this study (Ullah et al., 2025).

Furthermore, attaining generalisability continues to be a primary goal. A model that excels on a particular dataset may not consistently uphold its effectiveness when utilised with new patients or in other clinical environments. The absence of generalisation restricts the practical utility of deep learning models in real-world healthcare settings. Therefore, it is essential to create models that are both precise and durable, efficient at managing varied data situations.

In addition, even if the suggested segmentation framework might perform well within the chosen dataset, it cannot be immediately applied to larger clinical environments without

additional validation and further modification. Hospitals, scanners, operators, acquisition procedures, and patient groups can all produce different echocardiographic pictures. So, the CAMUS dataset and the controlled experimental settings employed in this investigation are the limits of this solution. Before the model could be considered appropriate for actual clinical deployment, more external testing, clinical validation, and adaptation would be necessary.

1.2 Motivation

This research is motivated by the growing demand for automated and reliable resources in cardiac image analysis (Esteva et al., 2019). In clinical practice, manual segmentation of echocardiographic images is both time-intensive and prone to inter-observer variability, potentially impacting the consistency of diagnostic results. Consequently, there is an increasing demand for computational techniques that can aid clinicians by delivering precise and reproducible segmentation outcomes.

Recent progress in deep learning, especially convolutional neural networks, has shown considerable promise in medical image segmentation tasks. Architectures like U-Net and its variants are extensively utilised for their capacity to capture spatial and contextual features, making them highly effective for the analysis of complex medical images. The effectiveness of these models is affected by various factors, including loss functions, data sampling methodologies, and training configurations. This requires a systematic examination of how various design decisions influence segmentation accuracy and robustness. Recent research highlighted the significance of architectural selection, preprocessing accuracy, and robust benchmarking on CAMUS, especially in the comparative analysis of U-Net, Attention U-Net, and contemporary hybrid models within standardised conditions (Ullah et al., 2025).

A primary motivation for this study is the difficulty of attaining generalisation. Models trained on particular datasets might display inconsistent performance when applied to new patients or different imaging conditions. Consequently, it is essential to investigate methodologies such as suitable data partitioning and refined training techniques to augment model generalisability and guarantee reliable performance on unique data.

Besides segmentation accuracy, it is essential to correlate model outputs with clinically significant outcomes. This study primarily emphasises segmentation; however, the predicted masks can also be utilised to estimate critical cardiac parameters, including end-diastolic

volume (EDV), end-systolic volume (ESV), and ejection fraction (EF). These parameters offer significant insight into cardiac function and illustrate the potential practical utility of automated segmentation techniques. It is essential to recognise that these estimates originate from segmentation outputs based on simplified assumptions and are meant to offer indicative rather than clinically validated measurements.

This research aims to establish a robust and generalisable deep learning framework for cardiac structure segmentation, while investigating its potential for clinically relevant analysis.

1.3 Research Questions

This study aims to address the challenges outlined in the problem statement by answering the following research questions:

- What is the impact of various loss functions on the effectiveness of deep learning models in cardiac structure segmentation?
- What is the effect of different sampling strategies on addressing class imbalance in echocardiographic data?
- To what degree do architectural enhancements, including attention mechanisms and sophisticated U-Net variants, improve segmentation accuracy?
- In what ways can generalisation be enhanced through effective data partitioning and training techniques?

The research questions direct the experimental design and establish a systematic framework for assessing the effectiveness of various approaches.

1.4 Aim and Objectives

The main objective of this project is to create generalisable deep learning models for the segmentation of cardiac structures utilising echocardiographic images.

- To attain this objective, the subsequent aims are defined as follows:
- To execute and assess various deep learning architectures, including U-Net, Attention U-Net, and U-Net++.

- To examine the impact of various loss functions, including cross-entropy, Dice loss, and focal loss, on segmentation efficiency.
- To evaluate the impact of data sampling methodologies, encompassing weighted, balanced and improved sampling, in mitigating class imbalance.
- To guarantee dependable model assessment through the application of patient-level data partitioning to avert data leakage.
- To evaluate model performance utilising quantitative metrics, including the Dice coefficient and Hausdorff Distance (HD95).
- To evaluate clinically significant parameters, including end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF), derived from segmentation results.

1.5 Scope of Study

This research examines the use of deep learning methodologies for the segmentation of cardiac structures in echocardiographic images. The experiments utilise the publicly accessible CAMUS dataset to ensure reproducibility and consistency. The project involves data preprocessing, patient-level data segmentation, model development, training, evaluation, and clinical metric estimation.

This study focuses exclusively on 2D echocardiographic segmentation of cardiac structures, specifically the left ventricle, myocardium, and left atrium. A new clinical system is not suggested by this study. Rather, it evaluates current model architectures and enhances training via loss functions and sampling methodologies.

The study is constrained by the size of the dataset, processing power, and the inherent unpredictability of echocardiographic data. Although the results offer insightful information, they are restricted to the dataset that was used and might need additional testing, validation, and modification in larger clinical settings.

1.6 Structure of The Report

This report is divided into six chapters to provide an ordered and systematic presentation of the study. Chapter 1 presents the study's background, outlines the problem statement, research

questions, aims and objectives, and specifies the project's scope. Chapter 2 offers a detailed review of the current literature regarding cardiac image segmentation, deep learning models and loss functions, data balancing methodologies, and useful clinical applications. Chapter 3 defines the methodology used during this research, addressing dataset preparation, preprocessing, data segmentation, model architectures, training strategies, post-processing approaches, assessment metrics, and implementation details. Chapter 4 presents the experimental data and includes an initial discussion using quantitative and qualitative analysis, comparisons with current research, and interpretation of important results. Chapter 5 provides an in-depth study of the results obtained, critically evaluating the influence of various design decisions and comparing the findings with existing research. Chapter 6 ultimately concludes the study by summarising the main contributions, outlining the limits, and suggesting directions for future research.

CHAPTER 2: LITERATURE REVIEW

Cardiac image segmentation is an important part of medical image analysis, particularly in echocardiography, where accurate identification of cardiac structures supports diagnosis, treatment planning, and the estimation of clinical measures such as end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF). Echocardiography is widely used because it is non-invasive, relatively low-cost, and provides real-time visualisation of heart structure and function. However, automated segmentation is challenging because echocardiographic images often contain speckle noise, low contrast, artefacts, and patient-dependent variation. Conventional methods have relied on manual annotation or traditional image processing techniques, but these approaches are limited in terms of generalisability, consistency, and scalability. Deep learning has therefore become increasingly important in medical imaging because it can learn relevant image features directly from data (Litjens et al., 2017). Systematic reviews also show that deep learning systems can perform comparably to healthcare professionals in selected imaging tasks, although outcomes vary across datasets and clinical settings (Liu et al., 2019; Esteva et al., 2019).

Convolutional neural networks (CNNs) have played an important role in medical image segmentation. U-Net is one of the most widely used architectures because of its encoder-decoder structure and skip connections, which help preserve spatial information during segmentation (Ronneberger et al., 2015). U-Net-based models have demonstrated strong performance in cardiac segmentation, including on benchmark datasets such as CAMUS (Leclerc et al., 2019). Several U-Net variants have also been developed to improve segmentation quality. Attention U-Net uses attention gates to focus on relevant image regions and reduce background influence, which is useful in noisy and low-contrast images (Oktay et al., 2018). U-Net++ uses nested skip connections to improve feature propagation and reduce the semantic gap between encoder and decoder features (Zhou et al., 2018). More recent hybrid architectures, such as TransUNet, combine CNN feature extraction with transformer-based self-attention to capture global context, although they often require greater computational resources (Chen et al., 2021). Recent CAMUS-focused benchmarking also indicates that architectural complexity alone does not guarantee improved outcomes, as performance depends on preprocessing, training strategy, and evaluation design as well as the model architecture (Ullah et al., 2025).

Loss-function design is another important factor affecting segmentation performance. Although standard cross-entropy loss is commonly used, it may be less effective when foreground structures occupy a smaller area than the background. Dice loss is suitable for segmentation tasks because it directly optimises the overlap between predicted and ground-truth masks (Milletari et al., 2016). Focal loss addresses class imbalance by reducing the contribution of easily classified examples and encouraging the model to focus on more difficult samples (Lin et al., 2017). Class-balanced optimisation has also been proposed to give greater weight to under-represented classes based on effective sample size (Cui et al., 2019). These methods support the use of hybrid and class-aware loss functions in cardiac segmentation, where smaller structures such as the myocardium and left atrium can be more difficult to segment accurately.

Data-level techniques such as balanced and weighted sampling can also be used to address class imbalance. In segmentation tasks, the background class often dominates the pixel distribution, which may bias the model towards majority-class predictions. Sampling techniques increase the representation of minority structures during training and have been shown to improve learning in imbalanced datasets (Buda et al., 2018). This is especially relevant for echocardiographic segmentation because clinically important structures occupy a smaller percentage of the image than the background. However, sampling strategies must be evaluated together with loss functions and model architecture, as segmentation performance depends on the interaction between these design choices.

In 2D echocardiography, the CAMUS dataset is now a widely used benchmark for cardiac segmentation. It contains apical two-chamber and four-chamber images with expert annotations for the left ventricle, myocardium, and left atrium (Leclerc et al., 2019). Its variation in image quality, patient anatomy, and acquisition conditions makes it appropriate for assessing model robustness and generalisation. Previous research also highlights the importance of patient-level data splitting to prevent data leakage, because using images from the same patient in both training and testing sets can produce overly optimistic results (Leclerc et al., 2019).

Several metrics are required to evaluate segmentation models effectively. The Dice coefficient is commonly used to measure overlap between predicted and ground-truth masks, while Intersection over Union (IoU) provides an additional region-based similarity measure. The 95th percentile Hausdorff Distance (HD95) is frequently used to evaluate boundary accuracy, which

is important because even small boundary errors may affect clinical interpretation (Leclerc et al., 2019). In addition to segmentation accuracy, predicted masks can be used to estimate clinical indicators such as EDV, ESV, and EF. However, these estimates should be interpreted cautiously unless verified against clinical ground truth, as they often rely on simplified assumptions.

Overall, the literature indicates that deep learning has significantly advanced cardiac image segmentation, yet several challenges remain. Many studies focus mainly on improving Dice scores, while fewer studies systematically investigate the combined impact of architecture, loss functions, sampling strategies, leakage prevention, and clinically relevant metric estimation. There is an opportunity to investigate the interaction between weighted, balanced, and improved sampling strategies and hybrid loss functions under a consistent patient-level splitting framework. Robustness and generalisation remain particularly important in echocardiography because image quality and anatomical appearance can vary substantially between patients and imaging conditions.

2.1 Research Gap

Although U-Net and its variants have demonstrated strong performance in cardiac segmentation, there remains a need for careful evaluation of how model architecture, loss functions, and sampling strategies interact under a controlled training framework. In particular, the relationship between loss-based methods such as Dice loss and focal loss, and data-level methods such as weighted, balanced, and improved sampling, remains underexplored in echocardiographic segmentation. Inadequate patient-level data splitting can also lead to data leakage and overly optimistic performance estimates.

Recent studies have also highlighted the importance of robustness, quality control, and clinically meaningful outputs. For example, model-agreement and graph-based refinement methods have been explored to improve cardiac segmentation reliability and support ejection fraction estimation (Van de Vyver et al., 2024). However, limited research combines segmentation performance, class imbalance handling, leakage prevention, and clinical metric estimation within a single reproducible framework. This study aims to address these gaps by comparing multiple deep learning architectures, analysing different loss and sampling strategies, applying patient-level data splitting, and extending the analysis to clinically relevant metrics derived from segmentation outputs. This study does not introduce a new network

architecture. Rather, it is designed to evaluate the existing designs under controlled conditions, and improve segmentation performance through loss-function design, sampling strategies and leakage-aware evaluation.

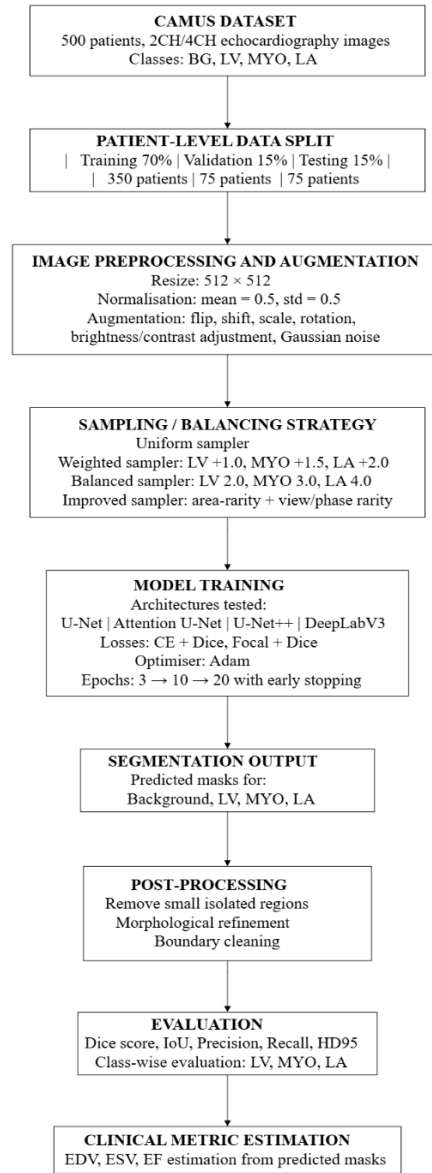
CHAPTER 3: METHODOLOGY

3.1 Overview

This chapter outlines the methodology used to create a generalisable deep learning framework for heart structure segmentation using echocardiographic images. The whole pipeline includes preparing the dataset, preprocessing it, building the model, training it, post-processing it, and testing it. The main goal is to accurately separate cardiac structures while making sure that the model is strong and can work with data from new patients.

Deep learning, especially convolutional neural networks (CNNs), has made a big difference in medical image segmentation by making it possible to automatically learn hierarchical spatial characteristics (Litjens et al., 2017). This study examines several architectures and training methodologies to comprehensively assess their influence on segmentation performance. Figure 1 presents the complete deep learning pipeline used in this study, including dataset preparation, patient-level splitting, preprocessing, sampling strategies, model training, post-processing, evaluation, and clinical metric estimation

Figure 1 Overall architecture of the proposed cardiac echocardiography segmentation pipeline



3.2 Dataset Description

This research uses the CAMUS dataset, a publicly available benchmark specifically designed for cardiac ultrasound segmentation tasks (Leclerc et al., 2019). The dataset for this research was obtained from the Kaggle CAMUS human heart data repository (Hasan, n.d.). The dataset can be accessed at: <https://www.kaggle.com/datasets/shoybhasan/camus-human-heart-data>

The collection contains echocardiographic images collected from multiple patients, including both apical two-chamber (2CH) and four-chamber (4CH) views. Each image includes expert annotations for clinically important cardiac structures, including the left ventricle (LV), myocardium (MYO), and left atrium (LA). The dataset has been used in research examining

limited-data learning and representation transfer, making it suitable for evaluating segmentation accuracy and generalisation under constrained supervision (Saeed et al., 2022).

Echocardiographic imaging poses numerous obstacles in comparison to other modalities like MRI or CT. These include speckle noise, low contrast between structures, shadowing effects, and a lot of differences in anatomy between patients. These characteristics make it hard to get the right segmentation; therefore, strong deep learning methods are needed.

Prior studies have shown that temporal consistency and stable preprocessing are important for reliable echocardiography segmentation performance (Painchaud et al., 2022). So, it is important to handle the dataset carefully to make sure that model training and testing are accurate.

3.3 Data Preprocessing and Augmentation

Data preprocessing is an important stage in the preparation of datasets for deep learning models. In this study, all images are resized to a uniform spatial resolution of 512×512 pixels to provide consistency across the dataset. Intensity normalisation is applied to standardise pixel values, improving training stability and convergence.

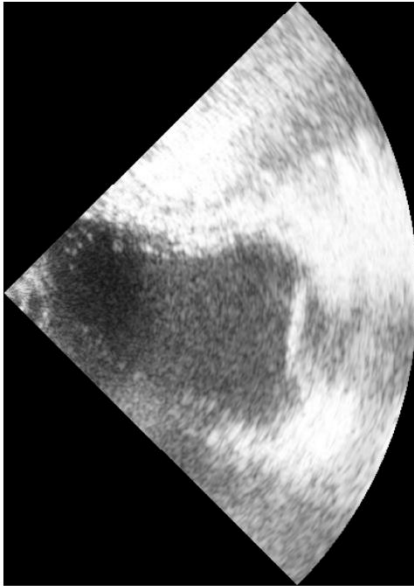
Normalisation reduces inconsistencies caused by varying imaging circumstances and improves the model's ability to recognise important features. Consistent preprocessing helps maintain stable input distributions during training, which is important for reliable echocardiography segmentation (Painchaud et al., 2022). Recent benchmarking of CAMUS has shown that preserving preprocessing integrity and assuring alignment and intensity consistency substantially influence downstream segmentation accuracy, highlighting the importance of a careful preprocessing pipeline in this study (Ullah et al., 2025).

As a visual demonstration of the preprocessing procedure, an example raw echocardiographic image and its pre-processed form is shown in Figure 2. This comparison shows how the source image is scaled and normalised before being fed into the model. This graphic illustrates the intermediate step of preprocessing that happens before training and is not the final prediction output.

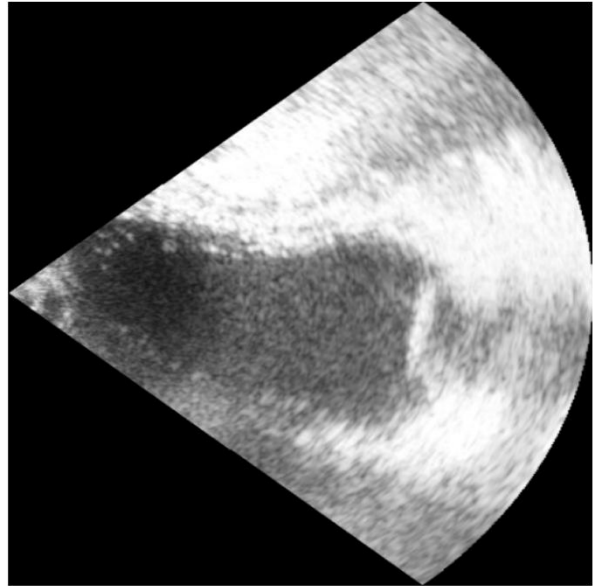
Figure 2 Examples of raw and preprocessed echocardiographic images before model training.

Examples of Raw and Preprocessed Echocardiographic Images

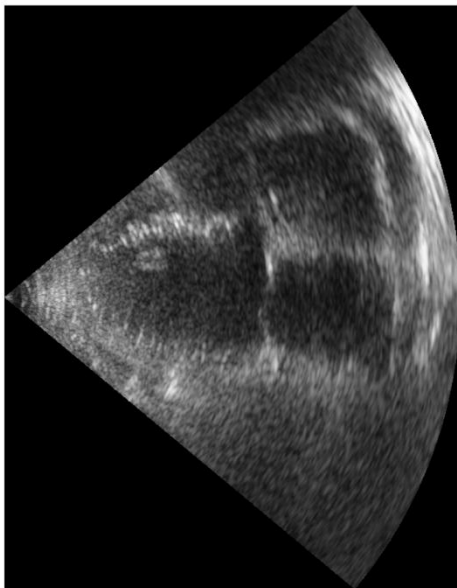
Raw Image 1



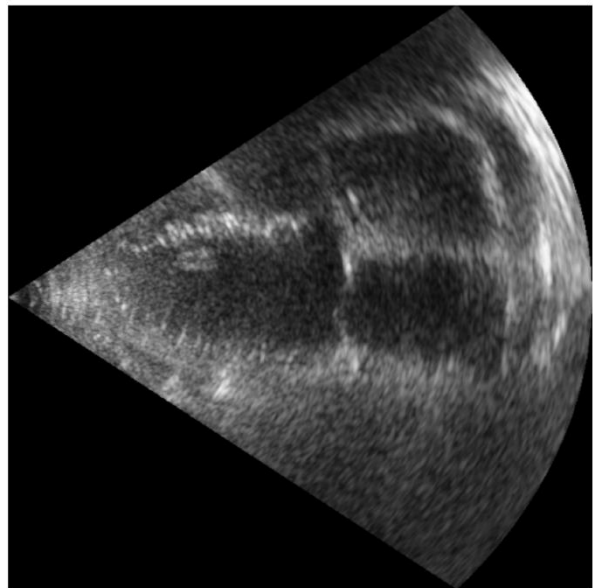
Preprocessed Image 1



Raw Image 2



Preprocessed Image 2



Data augmentation techniques are used to improve model generalisation. This includes geometric manipulations like horizontal flipping and affine transformations (including minor shifts, scaling, and rotations), alongside intensity-based augmentations such as brightness and contrast modifications and the incorporation of Gaussian noise. These changes replicate real-world variability in echocardiographic images, encompassing variations in acquisition settings and noise levels.

In medical imaging, where annotated datasets are often limited, augmentation is crucial for improving model robustness (Shorten and Khoshgoftaar, 2019). Augmentation enhances the model's capacity to generalise to unfamiliar data and mitigates the likelihood of overfitting by exposing it to a wider range of variances that occur during training.

3.4 Data Splitting and Leakage Prevention

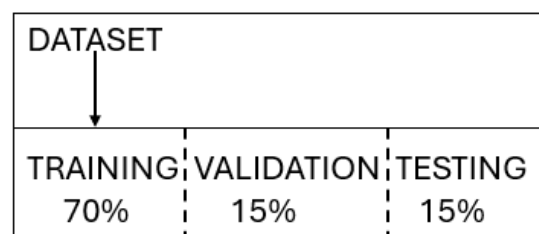
The dataset is split into training, validation, and testing subsets utilising a patient-level splitting approach. This guarantees that images from the same patient are excluded from multiple subsets, thus averting data leakage.

Data leakage presents a substantial issue in medical imaging, perhaps leading to excessively favourable performance assessments. When images from the same patient are included in both training and testing datasets, the model may memorise patient-specific characteristics rather than acquiring generalisable patterns.

This study utilised a dataset partitioned in a 70:15:15 ratio for training, validation, and testing, respectively. This yielded 350 patients for training, 75 for validation, and 75 for testing, totalling 500 patients. The respective quantities of image samples were around 1400 for training, 300 for validation, and 300 for testing.

The patient-level division ensures that the model is assessed on completely unique data, offering a more accurate evaluation of its generalisation capacity. This methodology is strongly supported in medical imaging research (Leclerc et al., 2019). Figure 3 depicts the dataset splitting approach utilised in this research.

Figure 3 Patient-level dataset split showing training (70%), validation (15%), and testing (15%) partitions



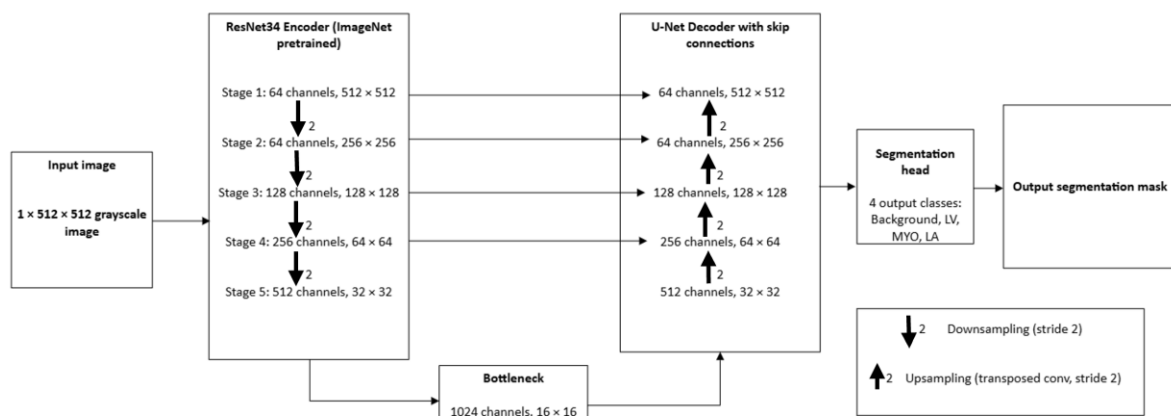
3.5 Model Architectures

In this study, different deep learning architectures are used to assess their performance at segmenting cardiac images. By using different architectures, we can compare different design choices and assess their effect on the accuracy of segmentation and the performance of generalisation. The architectures were used in their existing implemented forms. The main contribution of this study is a systematic comparison of these models along with improvements in training, such as loss functions, sampling strategies, patient-level partitioning, and post-processing techniques.

The U-Net architecture is used as a starting point because it has been shown to work well in medical image segmentation tasks (Ronneberger et al., 2015). The U-Net uses an encoder-decoder architecture. The encoder uses convolutional and pooling layers to capture hierarchical features, and the decoder uses up sampling operations to reconstruct spatial resolution. A key advantage of U-Net is that it uses skip connections to send detailed spatial information from the encoder to the decoder. This improves localisation accuracy and helps preserve structural information.

The segmentation-models-pytorch library and a ResNet34 encoder were used to build the U-Net baseline for this study. The encoder used weights that had already been trained on ImageNet. It had one input channel for greyscale echocardiographic images and four output classes that stood for background, left ventricle, myocardium, and left atrium. Therefore, the implemented U-Net follows an encoder-decoder structure with skip connections, and ResNet34 is used as the backbone for feature extraction. The U-Net architecture that was used in this study is shown in Figure 4.

Figure 4 Implemented U-Net architecture used in this study. The model uses a ResNet34 ImageNet-pretrained encoder, a U-Net decoder with skip connections, one grayscale input channel, and four output classes: background, LV, MYO, and LA.



Attention U-Net is an improved version of the baseline model. It uses attention-based feature refinement to help the network focus on important parts of the image and ignore irrelevant background information (Oktay et al., 2018). In echocardiographic images, this is especially helpful because noise and artefacts can hide important anatomical structures. Attention mechanisms improve the accuracy of segmentation by drawing attention to specific features, especially in small, complicated areas like the myocardium. This decision is further supported by recent CAMUS benchmarking studies, which revealed that Attention U-Net yields moderate yet significant enhancements in low-contrast and small-structure areas by mitigating boundary leakage compared to simpler baselines (Ullah et al., 2025).

This study also evaluates DeepLabV3 to see how well it can gather multi-scale context data using atrous/dilated convolution (Chen et al., 2017). Atrous convolution lets the model increase the receptive field without making the computations more difficult. This makes it easier to separate structures of different sizes. This feature is especially helpful for cardiac imaging, where anatomical parts are very different in size.

Also, U-Net++ is being investigated to improve the efficiency of segmentation. The semantic difference between encoder and decoder feature maps is reduced by U-Net++'s nested and dense skip connections (Zhou et al., 2018). In comparison to the standard U-Net architecture, this leads to better feature propagation and more accurate segmentation boundaries.

A U-Net configuration based on residual encoders is also looked at because residual learning might make gradient flow and training stability better. Residual connections fix the problem of vanishing gradients and make it easier to train deeper networks, which leads to better performance and convergence.

Recent research shows that transformer-based architectures like TransUNet could improve global-context modelling and generalisation in difficult echocardiographic frames (Ullah et al., 2025) but they weren't used in this study.

Therefore, the architectural comparison is performed along with the sampling strategies and loss functions, since the final segmentation performance depends on how these parts work together. This research uses several different architectures to fully evaluate a wide range of design options. These architectures include feature extraction methods, attention techniques, and multi-scale context modelling. This method for comparing makes it easier to find the best architecture for cardiac segmentation tasks in a variety of settings. Table 1 defines the principal

architectural configurations evaluated in this study, highlighting their essential characteristics and benefits for cardiac image segmentation.

Table 1 Model Architecture Comparison

MODEL	KEY FEATURE	ADVANTAGE
U-Net with ResNet34 encoder	Encoder-decoder with skip connections	Strong baseline with pretrained feature extraction
ResUNet-style/residual encoder U-Net	Residual encoder learning	Improved convergence
DeepLabV3	Atrous /dilated convolution	Multi-scale features
Attention U-Net	Attention-based feature refinement	Focus on relevant regions
U-Net++	Dense skip connections	Improved feature propagation

3.6 Loss Functions

Loss functions are crucial in directing the optimisation process of deep learning models, especially in segmentation tasks where class imbalance poses a considerable barrier. This work investigates several loss functions to enhance segmentation accuracy and robustness.

The Dice coefficient is used as a fundamental metric for assessing segmentation performance. It measures the intersection between the predicted segmentation mask and the ground truth mask, making it particularly appropriate for medical imaging applications that require accurate border identification. The Dice coefficient is defined as follows:

$$Dice = \frac{2 | P \cap G |}{| P | + | G |}$$

P indicates the predicted mask, while G represents the ground truth mask. An elevated Dice score signifies superior overlap and segmentation quality.

In addition to Dice, Intersection over Union (IoU) is utilised as a complementary evaluation metric to assess region similarity:

$$IoU = \frac{| P \cap G |}{| P \cup G |}$$

Although Dice loss directly optimises segmentation overlap, it may inadequately resolve the problem of class imbalance. In echocardiographic images, the background class dominates; potentially resulting in biased learning whereby the model favours background prediction over smaller anatomical structures.

Focal loss is integrated into the training process to reduce this issue. Focal loss emphasises challenging samples by minimising the influence of well-classified occurrences, therefore enhancing performance on minority classes (Lin et al., 2017). This is especially advantageous for distinguishing small structures like the myocardium.

This study employs a hybrid loss function that integrates Dice loss and focal loss. This combination guarantees that the model simultaneously optimises for overlap precision and class equilibrium. Prior research has shown that the integration of region-based and pixel-wise loss functions enhances segmentation performance (Milletari et al., 2016; Lin et al., 2017). To further address class imbalance beyond the loss formulation, class weighting was implemented during optimisation. The background class was assigned a weight of 0.2, while greater importance was attributed to clinically important structures, with weights of 1.0 for the left ventricle and myocardium, and 1.3 for the left atrium. This weighting method diminishes the influence of background pixels during training and promotes the model's emphasis on anatomically relevant areas, thereby improving overall segmentation performance.

3.7 Sampling Strategies

Medical image segmentation is affected by the class imbalance problem as the total number of pixels for the background class is usually much larger than other classes. This imbalance may negatively impact the model learning by biasing the prediction towards the majority background class. Uniform sampling reflects the natural distribution of the dataset. However, uniform sampling was not used as a key final experimental technique because it does not address class imbalance directly. To solve this challenge, this work applies and assesses weighted, balanced and improved sampling strategies.

3.7.1 Weighted Sampling

The weighted sampling puts additional weights on the samples with under-represented anatomical structures, which allows the model to learn from more clinically significant regions.

In the implementation, weighted sampling was used to prioritise samples having clinically significant structures. Additional weights were provided to the left atrium (+2.0), myocardium (+1.5) and left ventricle (+1.0).

3.7.2 Balanced Sampling

Balanced sampling improves this idea by modifying the sample selection process to promote a more balanced representation of cardiac structures throughout training. In the implementation, we used balanced sampling with set weights of 4.0, 3.0 and 2.0 for left atrium, myocardium and left ventricle correspondingly, whereas a weight of 1.0 was used for background samples.

3.7.3 Improved Sampler

Weighted and balanced sampling were studied as well as an improved sampler, which is a customised adaptive sampling approach. Unlike fixed sampling techniques, the improved sampler prefers training samples with clinically relevant foreground structures, especially small or hard to segment regions like the left atrium and myocardium. Moreover, it considers contextual rarity such as infrequent image views or cardiac phases to further expose the model to more diverse and informative examples. This sampler is designed for reducing background dominance and improving the learning of difficult anatomical structures.

The improved sampler dynamically weighted sample importance based on the size of the anatomical structure and contextual rarity. Area-based scaling was used to give higher weights to smaller structures, and under-represented imaging views and cardiac phases were also prioritised. Final weights were normalised to help in training stability.

Proper sampling methods can improve the performance of the model on imbalanced datasets (Buda et al., 2018). Hence, it is a systematic way for evaluating the impact of class balancing at data level on segmentation accuracy to compare weighted, balanced and improved sampling strategies.

In addition, in a study on intermediate class distribution, pixel proportions were estimated through batch-wise sampling. Note that this analysis was done on sampled batches, not full dataset counts so there may be some slight variation in the estimated class distributions across runs.

3.8 Training Strategy

A progressive epoch technique was implemented to provide consistent training and effective experimentation. Initial tests were performed utilising a limited number of epochs (e.g., 3 epochs) to verify the accuracy of the implementation and training pipeline. Subsequently, the number of epochs was incrementally increased (e.g., 10 and then 20 epochs) to enable the models to learn more complex feature representations. Models were trained for a maximum of 20 epochs during the last experimental phase, though some models stopped before completing the entire epoch count due to early stopping. Early stopping was utilised during this process to automatically stop training when validation performance no longer improved, thus preventing overfitting and ensuring appropriate model generalisation.

The Adam optimiser is used for model training because of its flexible learning rate features and effective convergence characteristics (Kingma and Ba, 2015). Adam integrates the benefits of momentum and adaptive learning rates, making it highly effective for complicated deep learning tasks. This study uses a traditional supervised training pipeline; however, previous research on CAMUS and EchoNet-Dynamic suggests that representation learning via contrastive pretraining can enhance segmentation with limited annotations, particularly for DeepLabV3 and U-Net architectures (Saeed et al., 2022).

Learning rate scheduling is implemented to progressively decrease the learning rate throughout the training process. This enables the model to implement substantial modifications during the initial phases of training and more refined improvements as it reaches convergence. This method enhances stability and prevents overshooting optimal solutions.

Extra training parameters, such as batch size and epoch count, are selected by empirical observations. These parameters are calibrated to optimise computational efficiency and model performance.

3.9 Post Processing

Because the predicted masks may still contain small inconsistencies and noise, post-processing techniques are used to improve segmentation results. In this study, minor isolated areas that are not anatomically significant are eliminated, and morphological operations are used to maintain continuity in segmented regions and refine borders. Particularly in echocardiographic images,

where CNN outputs may produce noisy or anatomically inconsistent borders under suboptimal image quality conditions, these techniques help reduce false positives and improve the segmentation's visual quality (Leclerc et al., 2019; Kim et al., 2021).

3.10 Evaluation Metrics

The evaluation of model performance uses many quantitative measures to deliver a thorough assessment of segmentation quality. Each indicator captures a distinct aspect of model performance, guaranteeing a comprehensive evaluation.

Table 2 Evaluation Metrics

Metric	Purpose
Dice Score	Overlap accuracy
IoU	Region similarity
Precision	False positive control
Recall	False negative control
HD95	Boundary accuracy

The Dice score is used as the main statistic in this investigation, as it directly measures segmentation overlap. IoU offers more understanding of region similarity, whereas recall and precision assess classification performance at the pixel level.

The 95th percentile Hausdorff Distance (HD95) evaluates border precision, crucial in medical imaging for the exact definition of anatomical structures. Using various indicators guarantees the evaluation of both regional and boundary performance.

3.11 Clinical Metrics Estimation

Besides segmentation accuracy, the model's clinical significance is evaluated by estimating essential cardiac parameters from the predicted segmentation masks. These include End-Diastolic Volume (EDV), End-Systolic Volume (ESV), and Ejection Fraction (EF).

The Ejection Fraction is determined by the following formula:

$$EF = \frac{EDV - ESV}{EDV} \times 100$$

These measures offer significant insights into heart function and are extensively used in clinical diagnosis. These are however estimates based on simplified assumptions and should be taken as indicative measures and not clinically validated measurements.

Prior studies have emphasised the significance of integrating segmentation results with clinical metrics to enhance the practical use of deep learning models (Leclerc et al., 2019).

3.12 Generalisation Strategy

An important goal in this study is to ensure model generalisation. If a model performs well on the training data but fails to generalise on unseen data it is not suitable for practical applications. This problem is solved using patient-level data partitioning to prevent data leakage so that the model is tested on completely unseen patients. Data augmentation is used to increase variability. Early stopping and learning rate scheduling are used to avoid overfitting and to ensure stable performance on new unseen data.

Generalisation is a major challenge for medical AI systems and strict evaluation procedures are essential to ensure clinical reliability (Liu et al., 2019).

3.13 Limitations

Despite this comprehensive methodology, numerous restrictions exist:

- This study depends on a singular dataset, perhaps restricting the applicability of the results to different clinical environments.
- Differences in imaging methods, patient demographics, scanner settings, and acquisition conditions may influence model performance.
- The estimate of clinical measures relies on simplified assumptions and may not accurately reflect clinical measuring methodologies. Although these estimates offer valuable information, they must be used carefully.
- This study does not address developing architectures, such as transformer-based models, due to computational limitations. These models have demonstrated

encouraging outcomes in recent studies and may present opportunities for future improvement.

- The proposed framework is evaluated under controlled experimental conditions and would require further validation and possible modification before use in wider clinical environments.

3.14 Implementation Details

This study employed deep learning models utilising the Python programming language. The primary framework used for model development and training was PyTorch, which provides adaptability and efficient GPU acceleration for deep learning tasks. Libraries including NumPy and Pandas were applied for data manipulation and preprocessing, whereas Matplotlib and Seaborn were used for result visualisation.

The experiments were performed in the Google Colab environment utilising an NVIDIA Tesla T4 GPU, facilitating the effective training of convolutional neural networks. GPU acceleration greatly decreased training duration and enabled testing with various architectures, loss functions, sampling strategies, and hyperparameter settings.

All experiments were performed under controlled and reproducible conditions. A fixed random seed of 42 was utilised throughout Python, NumPy, and PyTorch to enhance reproducibility. However, slight differences in outcomes may still happen between runs because of stochastic elements in the training process, such as data augmentation, sampling techniques, and GPU-based computation.

The input images were resized to 512×512 pixels spatial resolution and batch size was set to 2 due to memory constraints of high-resolution medical images. The models were trained for up to 20 epochs with early stopping used to terminate training if no improvement was observed on the validation performance. The whole pipeline included data loading, pre-processing, augmentation, model training, validation, post-processing and evaluation with segmentation and clinical metrics.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Overview

This chapter outlines the experimental findings and provides an in-depth analysis of the performance of the deep learning models created for cardiac structure segmentation utilising echocardiographic images (Litjens et al., 2017). The results are examined through quantitative metrics and qualitative visualisations to deliver a thorough assessment of model behaviour and segmentation quality. The experiments were performed iteratively from Week 1 to Week 7, integrating incremental enhancements in model architecture, loss functions, sampling strategies, and training settings.

The evaluation focuses on segmentation accuracy and boundary precision, quantified using the Dice coefficient and the 95th percentile Hausdorff Distance (HD95) (Leclerc et al., 2019). Alongside overall performance, a class-wise examination is conducted for clinically significant cardiac structures, including the left ventricle (LV), myocardium (MYO), and left atrium (LA). The results are analysed in relation to model design choices and compared with existing literature to evaluate the performance and generalisability of the suggested approach.

4.2 Week-By-Week Performance Progression

Table 3 week-wise improvement summary

WEEK	KEY IMPROVEMENT	BEST DICE SCORE
Week 1	Baseline models	~0.86–0.87
Week 2	Introduction of loss functions and sampling	~0.87
Week 3-5	Hyperparameter tuning and refinement	~0.88
Week 6	Advanced architectures (U-Net++)	~0.89
Week 7	Final optimisation	0.899058

Values presented for Weeks 1-6 are approximate ranges of performance found through repeated experimentation. Only the final Week 7 result corresponds to the recorded and reproducible model output used for evaluation.

The results indicate a clear and consistent improvement in segmentation performance throughout the various experimental phases. The baseline models developed in Week 1 attained Dice scores between 0.86 and 0.87, reflecting satisfactory performance although constrained capacity to accurately capture complex anatomical details (Ronneberger et al., 2015). The implementation of sampling algorithms and enhanced loss functions in Week 2 yielded moderate performance improvements, underscoring the need to address class imbalance.

Subsequent enhancements from Weeks 3 to 5, incorporating modifications to training settings and hyperparameters, resulted in gradual advancements in model performance. Week 6 saw a notable enhancement with the implementation of advanced designs like U-Net++, which enabled superior feature propagation and augmented segmentation performance (Zhou et al., 2018). The concluding optimisation in Week 7 yielded a Dice score of 0.899058, illustrating the effectiveness of integrating many enhancements.

4.3 Final Model Configuration and Performance

The performance of the best-performing configurations across different architectures is summarised in Table 4.

Table 4 Performance Comparison of Selected Model Configurations

MODEL	SAMPLER	LOSS TYPE	MEAN DICE	HD95 (LV)
ATTENTIONUNET	weighted	focal_dice	0.899058	12.399
UNET BASELINE	weighted	focal_dice	0.897912	12.505
UNET++	balanced	focal_dice	0.892549	12.490
DEEPLABV3	improved	ce_dice	0.891949	13.159

Note: The left ventricle border measure (HD95_LV) is represented by the HD95 values shown in Table 4. Appendix A contains class-wise HD95 values for LV, MYO, and LA.

The expanded results table in Appendix A contains comprehensive performance measurements, such as ESV and EF estimate errors. The best-performing configuration is highlighted in bold to make the final selected model clear.

Table 4 emphasises the importance of optimising the overall training framework including model selection, sampling strategies, and loss functions, which is a critical tenet in deep

learning-driven medical image processing (Litjens et al., 2017). The best overall performance is achieved by the Attention U-Net model, indicating that attention mechanisms enhance the model's capacity to concentrate on relevant anatomical regions (Oktay et al., 2018). The best configuration in this study was the combination of Attention U-Net with weighted sampling and focal Dice loss, achieving a mean Dice score of 0.899058.

This finding is consistent with recent controlled benchmarking on the CAMUS dataset, which shows that regular U-Net continues to be a strong baseline while Attention U-Net improves segmentation performance in low-contrast regions and lowers boundary leakage (Ullah et al., 2025). Additionally, configurations using focal Dice loss combined with weighted sampling typically produce better results, underscoring the significance of both class balance and loss design in enhancing segmentation performance (Lin et al., 2017; Milletari et al., 2016).

4.4 Class-Wise Performance Analysis

Class-wise dice scores for the top-performing model are shown in Table 5 for additional segmentation performance analysis.

Table 5 Class-wise Dice Scores

STRUCTURE	DICE SCORE
LV	0.931915
MYO	0.868438
LA	0.896820

The best-performing Attention U-Net model's class-specific outputs show that the model achieves optimal accuracy for the left ventricle because of its larger anatomical size and more distinct border delineation in echocardiographic images. The myocardium, on the other hand, performs worse because of its intricate borders and thin structure, which make precise segmentation more difficult.

These results demonstrate an inherent challenge of segmenting smaller anatomical features and are consistent with previous studies showing differences in segmentation accuracy between various cardiac areas (Leclerc et al., 2019). These findings also show that anatomical structure

features have a significant impact on segmentation performance, with larger, well-defined sections being easier to segment than thin, low-contrast structures like the myocardium.

4.5 Impact of Sampling Strategies, Class Imbalance, and Loss Functions

Table 6 Pixel Distribution Across Sampling Methods

SAMPLER	BACKGROUND	LV	MYO	LA
UNIFORM	0.788105	0.078685	0.082331	0.050879
WEIGHTED	0.757455	0.085820	0.101553	0.055172
BALANCED	0.758341	0.091590	0.096387	0.053682
IMPROVED	0.754714	0.103500	0.097714	0.044071

Table 6 shows that the dataset is clearly unbalanced between classes, with the background class having most of the pixels. About 75% to 78% of the pixels are taken up by the background. Structures that are clinically important, like the left ventricle, myocardium, and left atrium, take up much smaller amounts (Buda et al., 2018). This difference makes segmentation tasks very hard because models tend to make predictions based on the background class that is most common, which makes it harder for them to learn accurately about minority classes. This finding fits with earlier research that said representation-balancing and class-sensitive optimisation are necessary in highly unbalanced segmentation settings, especially when clinically important structures only take up a small portion of pixels (Buda et al., 2018; Cui et al., 2019).

Several sampling strategies were tested during the experiment to find the best way to deal with this problem. These included a uniform, weighted, balanced, and an improved sampling approach. The natural distribution of the dataset is shown by uniform sampling, which was used for comparison in the class distribution analysis. However, it wasn't used as the main experimental strategy because it doesn't directly address class imbalance. To give the model more access to clinically important foreground structures, weighted, balanced, and improved sampling were used.

One important contribution of this research is an improved sampler. In contrast to fixed sampling methods, the improved sampler is designed to reduce the number of background pixels while increasing the visibility of clinically important structures. The improved sampler

raised the left ventricle's representation to about 0.103 and lowered the background proportion to about 0.754, as shown in Table 6. This suggests that the improved sampler gave more attention to clinically important foreground areas during training.

The improved sampler is more flexible than traditional weighted and balanced sampling methods because it takes into account both structural proportions and contextual variability. Weighted and balanced approaches are better at showing minority structures, but they mostly use static weighting schemes that might not fully show how complicated anatomical structures are. So, the improved sampling method is a good way to find out how to improve class representation (Buda et al., 2018; Cui et al., 2019).

It's important to look at the loss function along with the sampling strategy because segmentation performance is based on how well the model architecture, sampling strategy, and loss design work together, not just one of them. The choice of loss function is very important for the accuracy of the model. Combining dice loss and focal loss produced the best results out of all the configurations that were tested (Lin et al., 2017; Milletari et al., 2016). The Dice loss encourages the model to maximise the overlap between the predicted mask and the ground-truth mask. Focal loss, on the other hand, draws attention to difficult samples and lessens the effects of class imbalance.

However, the test results showed that weighted sampling along with focal Dice loss gave the best overall segmentation performance. This means that even though the improved sampler made the foreground representation better, the final best configuration was determined by how Attention U-Net, weighted sampling, and focal Dice loss worked together. These findings indicate that, in addition to increasing the representation of minority classes, addressing class imbalance requires considering their spatial characteristics and contextual distribution.

Note: Instead of using full dataset counts, the pixel shares were computed over sampled training batches. Because of this, there may be small differences between runs due to sampling differences between batches.

4.6 Qualitative Results (Visual Analysis)

Figure 5 Example segmentation results showing original images, ground truth masks, and predicted outputs

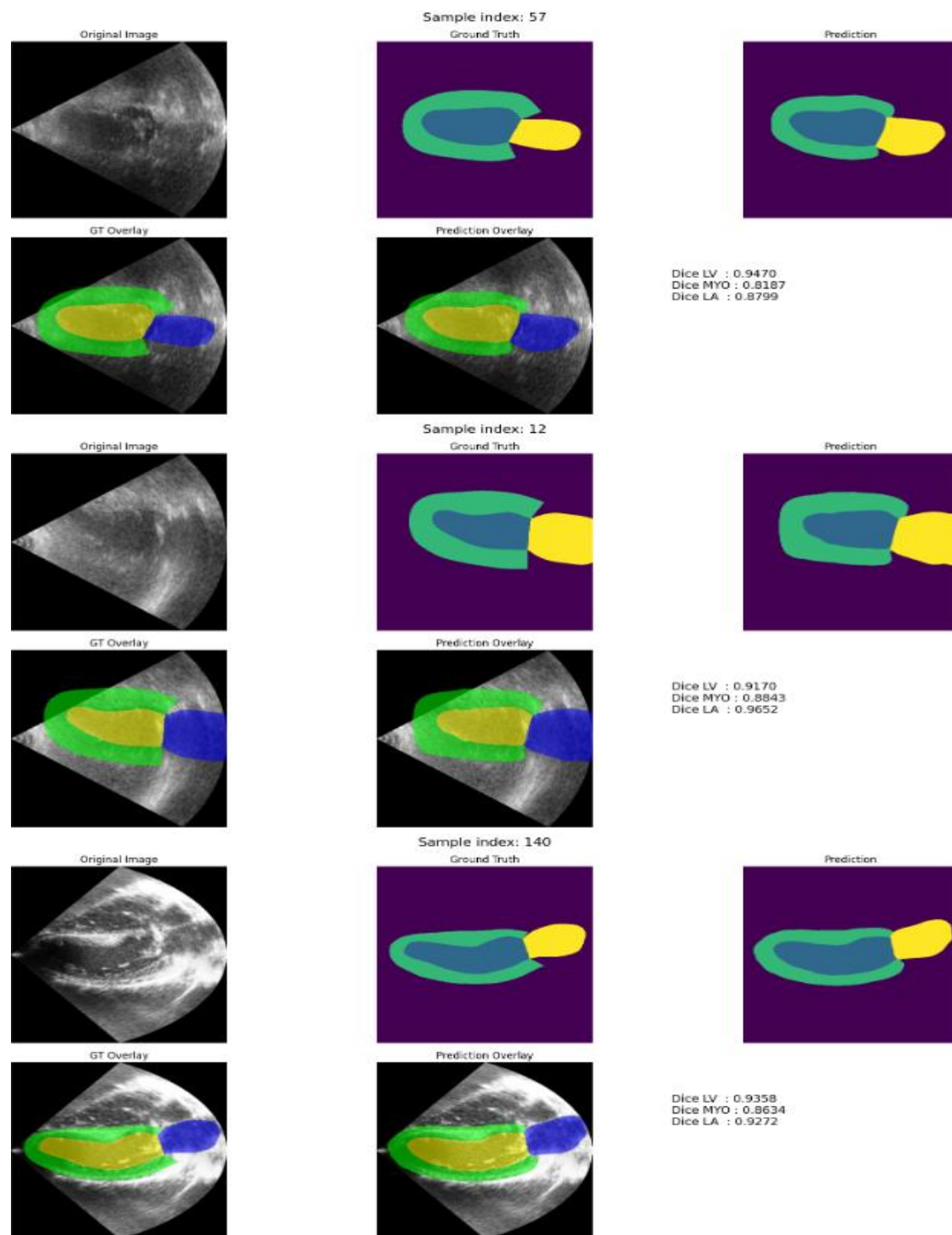


Figure 5 presents the qualitative results and illustrates the model's segmentation performance. The predicted segmentation masks match the ground truth annotations very well, especially for the left ventricle, where Dice scores are high across all samples shown. Even so, there are some small differences at the boundaries, especially in the myocardium, because echocardiographic images have noise and low contrast (Leclerc et al., 2019). Similar behaviour has been seen in

CNN-based left ventricle segmentation research; the outputs can still be used in clinical settings, but they may not be the same around thin or noisy myocardial borders, especially in harder echocardiographic frames (Kim et al., 2021).

The performance of the proposed training method is shown by the consistency seen across many samples. When focal Dice loss and sampling methods are added, they improve boundary localisation and help fix class imbalance, especially for smaller structures. By focusing on under-represented areas and rare anatomical configurations, the improved sampling method helps the model find complex structural elements that might be missed in natural or highly imbalanced training distributions.

Even though there are small boundary deviations, especially inside the myocardium, they don't have a big effect on the overall integrity of the anatomy or the clinical usefulness of the segmentation results. This means that the model strikes a good balance between being accurate and being useful for a wide range of situations. It gives results that are both competitive numerically and reliable visually across a wide range of echocardiographic samples.

4.7 Comparison with Existing Studies

Table 7 Comparison with Literature

STUDY	DICE SCORE
Ronneberger et al. (2015)	0.85–0.90
Leclerc et al. (2019)	~0.89
This Study	0.899058

The results of this investigation are comparable with existing studies, illustrating the impact of the proposed strategy. The final Dice score of 0.899058 indicates that the combination of an attention-based architecture, weighted sampling, and focal Dice loss achieved good segmentation performance. The results demonstrate that the proposed strategy achieves performance comparable to leading deep learning techniques in cardiac segmentation tasks (Litjens et al., 2017).

The extensive CAMUS research indicates that robust U-Net-based baselines frequently reach a performance plateau when data and preprocessing are adequately managed, implying that

improvements from complex architectures tend to be incremental rather than major (Leclerc et al., 2019; Ullah et al., 2025).

4.8 Generalisation Performance

Using patient-level data partitioning makes sure that the model is tested on patients that haven't been seen, which gives a more accurate picture of how well it works. There is good generalisation in the CAMUS dataset because the results are the same across several samples. However, more testing on outside datasets and various clinical settings is needed before it can be said for sure that the results are applicable to a wider range of patients (Liu et al., 2019).

4.9 Limitations

Despite strong performance, several limitations are present:

- The model is less accurate for thin structures such as the myocardium, mainly due to unclear boundaries, noise, and low contrast of echocardiography images.
- The reliance on a single dataset is a limitation to the broader generalisability of the results, which is a common limitation of clinical AI systems (Liu et al., 2019).
- The results should be interpreted considering the controlled setting of the CAMUS dataset and further validation is required before wider clinical application.

CHAPTER 5: DISCUSSION OF FINDINGS

5.1 Overview

This chapter presents a critical analysis of the experimental results discussed in Chapter 4. This section analyses the underlying reasons for the observed model behaviour, assesses the performance of various design choices, and correlates the findings with existing research in cardiac image segmentation, following the previous chapter's emphasis on reporting quantitative and qualitative results. The discussion highlights the influence of architectural selections, loss functions, sampling techniques, and training procedures on segmentation precision and generalisation performance.

5.2 Interpretation of Model Performance

The experimental results show that the segmentation performance increases gradually, and the final Dice score is 0.899058. This performance is comparable to the existing CAMUS-based studies (Leclerc et al., 2019), which means that the proposed methodology achieves competitive results.

The good performance of the Attention U-Net model suggests that attention mechanisms enable the model to focus on relevant anatomical regions in noisy and low-contrast echocardiographic images, resulting in improved segmentation quality (Oktay et al., 2018). This is in accordance with recent benchmarking studies which show that attention-based models can be effective for complex and small structures (Ullah et al., 2025).

But the relatively small differences in performance across the models suggest that architectural complexity alone is not enough to produce substantial improvement. Once a strong U-Net baseline is obtained, the next improvements heavily depend on optimisation techniques, data management and training settings. This is in line with previous studies which have suggested that performance improvements may stop when preprocessing and training pipelines are well optimised (Leclerc et al., 2019; Minaee et al., 2020).

5.3 Impact of Sampling Strategies and Loss Functions

Class imbalance is a recognised issue in medical image segmentation, where background pixels dominate the dataset and may bias model predictions (Buda et al., 2018). The findings indicate that sampling strategies influenced segmentation performance by increasing the representation of clinically important minority structures such as the myocardium and left atrium.

The improved sampling strategy provided a more adaptable approach by integrating structural proportions and contextual rarity. By prioritising smaller structures and under-represented imaging views or cardiac phases, it exposed the model to more informative and challenging samples. This aligns with class-balanced learning research, which highlights the importance of considering sample difficulty and distribution rather than relying only on class frequency (Cui et al., 2019).

However, the final results show that weighted sampling combined with focal Dice loss achieved the highest overall Dice score. Therefore, the improved sampler should be considered a useful exploratory method for enhancing class representation rather than the single best-performing strategy. This indicates that segmentation performance depends on the interaction between sampling method, loss function, and model architecture.

The combination of Dice loss and focal loss was important for attaining strong segmentation accuracy. Dice loss directly optimises overlap between predicted and ground-truth masks, while focal loss prioritises difficult-to-classify samples and reduces the impact of easily classified background pixels (Milletari et al., 2016; Lin et al., 2017). This balanced optimisation is especially useful in echocardiographic segmentation, where thin structures such as the myocardium are challenging to identify because of low contrast and noise. The findings therefore support previous research showing that hybrid loss functions can improve segmentation performance by addressing both class imbalance and structural complexity (Minaee et al., 2020).

5.4 Generalisation and Data Leakage Prevention

Model generalisation is a necessary requirement for clinical applications. In this work, we used patient-level data partitioning to prevent data leakage and to ensure that the model was tested on entirely unseen patients.

The consistency of the performance across the test samples indicates that the model has learned generalisable features within the CAMUS dataset, rather than memorising patient-specific patterns. Further generalisations would still need to be validated on external datasets and different clinical settings. The generalisation ability of medical AI systems is important as the performance of AI models can vary significantly due to different clinical tasks and settings (Liu et al., 2019).

The importance of correct partitioning of the data is well-known in the academic literature, as the failure to prevent data leakage can lead to overly optimistic performance estimates (Leclerc et al., 2019). The results of this work support the necessity of strict evaluation procedures in the development of trustworthy and clinically relevant models.

5.5 Clinical Relevance of Segmentation Results

In addition to the segmentation accuracy, the practical importance of the proposed method is that it supports clinical decision-making. The segmentation results can be used to compute important cardiac measures, such as end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF).

While the estimates are based on simplified geometric assumptions, they demonstrate the potential of automated segmentation systems to assist doctors in assessing heart function. However, they should be considered as indicative measures and not as clinically validated measurements. Segmentation reliability and temporal consistency were previously identified as major factors to improve the clinical usefulness of automated echocardiography analysis (Leclerc et al., 2019; Painchaud et al., 2022).

The high segmentation accuracy achieved in this study suggests that the proposed framework could be a useful basis for future clinically applicable systems, but further validation would be required before clinical use in real-world applications.

5.6 Limitations and Critical Reflection

Despite the achievement of competitive performance, various constraints need consideration:

- The study depends on a singular dataset, CAMUS, perhaps limiting the generalisability of the results to alternative clinical settings. Variations in imaging methods, patient populations, and equipment can substantially influence model performance.
- The segmentation of thin structures, such as the myocardium, presents difficulties, as seen by reduced Dice scores in comparison to bigger structures like the left ventricle. This highlights the underlying challenges of echocardiographic segmentation and indicates the necessity for enhancements in model architecture and training methodologies.
- Although newer architectures like transformer-based models have demonstrated potential in recent research, they were not investigated in this study due to computational limitations. Integrating such models may enhance the ability to obtain global contextual information (Chen et al., 2021; Ullah et al., 2025).
- The clinical metrics estimated from the predicted masks rely on simplified assumptions and therefore should be interpreted carefully rather than as clinically validated measurements.
- The proposed framework was evaluated under controlled experimental conditions and would require external validation and possible modification before real-world clinical deployment.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

The aim of this work was to develop a reliable and broadly applicable deep learning framework for automatically segmenting cardiac regions from echocardiographic pictures. To obtain clinically useful metrics like end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF), the left ventricle, myocardium, and left atrium must be precisely segmented. Reliable automated alternatives are needed because manual segmentation is labour-intensive and prone to inconsistency.

Several deep learning architectures, such as U-Net, Attention U-Net, U-Net++, and DeepLabV3, were evaluated in order to solve this issue. The study looked at how training configurations, sampling strategies, loss functions, and model selection affected segmentation performance. Additionally, patient-level data partitioning was used to minimise data leakage and facilitate a more trustworthy assessment of generalisation.

The results of the experiment demonstrate that the suggested framework performed competitively, with the optimal configuration achieving a final Dice score of 0.899058. The best overall performance was attained by Attention U-Net, suggesting that segmentation in noisy and low-contrast echocardiographic images can be supported by attention-based feature refinement.

The evaluation of an improved sampling strategy based on contextual rarity and anatomical structure size was another contribution of this study. Weighted sampling in conjunction with focal dice loss produced the best overall result, even though the improved sampler improved class representation. This demonstrates how architecture, sampling strategy, and loss-function design interact to determine segmentation performance.

Overall, this study shows that efficient training techniques, such as patient-level splitting, class-aware sampling, and hybrid loss functions, can enhance cardiac segmentation performance without exclusively depending on complex architectures. Although additional external validation and potential modification would be necessary prior to real-world clinical deployment, the suggested methodology offers a helpful foundation for future clinically relevant echocardiography segmentation systems.

6.2 Future Work

The proposed methodology exhibits strong performance; however, there are numerous opportunities for improvement and expansion:

- To increase generalisability beyond CAMUS, future studies should evaluate the proposed framework on multiple datasets with different patient demographics, imaging conditions, and clinical settings.
- More advanced data balancing strategies, such as curriculum-based learning or dynamically adjusting sampling methods during training, could be investigated.
- Innovative architectures using transformer models, such as TransUNet, may be explored to capture global contextual information and enhance segmentation precision in challenging scenarios with complex anatomical variations.
- The assessment of clinical parameters, such as EDV, ESV, and EF, could be enhanced through the application of more accurate volumetric modelling techniques and validation against clinical ground truth measurements.
- The model should be integrated into healthcare workflows and computational efficiency should be improved to investigate real-time implementation and deployment in clinical settings.
- To help find cases that need manual review, uncertainty estimation and quality assessment procedures could be combined.

REFERENCES

- Buda, M., Maki, A. and Mazurowski, M.A. (2018) ‘A systematic study of the class imbalance problem in convolutional neural networks’, *Neural Networks*, 106, pp. 249–259. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S0893608018302107?via%3Dihub>
- Chen, L.C., Papandreou, G., Schroff, F. and Adam, H. (2017) ‘Rethinking atrous convolution for semantic image segmentation’, *arXiv preprint arXiv:1706.05587*. Available at: <https://arxiv.org/abs/1706.05587>
- Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L. and Zhou, Y. (2021) ‘TransUNet: Transformers make strong encoders for medical image segmentation’, *arXiv preprint arXiv:2102.04306*. Available at: <https://arxiv.org/abs/2102.04306>
- Cui, Y., Jia, M., Lin, T.Y., Song, Y. and Belongie, S. (2019) ‘Class-balanced loss based on effective number of samples’, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9268–9277. Available at: <https://ieeexplore.ieee.org/document/8953804>
- Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S. and Dean, J. (2019) ‘A guide to deep learning in healthcare’, *Nature Medicine*, 25(1), pp. 24–29. Available at: <https://doi.org/10.1038/s41591-018-0316-z>
- Hasan, S. (n.d.) ‘CAMUS human heart data’, *Kaggle*. Available at: <https://www.kaggle.com/datasets/shoybhasan/camus-human-heart-data>
- Kim, T., Hedayat, M., Vaitkus, V.V., Belohlavek, M., Krishnamurthy, V. and Borazjani, I., 2021. Automatic segmentation of the left ventricle in echocardiographic images using convolutional neural networks. *Quantitative Imaging in Medicine and Surgery*, 11(5), p.1763. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8047352/>
- Kingma, D.P. and Ba, J., 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*. Available at: <https://arxiv.org/abs/1412.6980>
- Leclerc, S., Smistad, E., Pedrosa, J., Østvik, A., Cervenansky, F., Espinosa, F., Espeland, T., Berg, E.A.R., Jodoin, P.M., Grenier, T. and Lartizien, C., 2019. Deep learning for segmentation using an open large-scale dataset in 2D echocardiography. *IEEE transactions on medical imaging*, 38(9), pp.2198-2210. Available at: https://ieeexplore.ieee.org/abstract/document/8649738?casa_token=d4kwGXJZmAEAAAAA

[:TCgHmmLPShlgOo_dGKB0OpXotAkCUR2QgYtyQl_LIK6may6Ekd49Po76xsrpst6C3O
GEIQszQ](https://doi.org/10.1016/S2589-7500(19)30123-2)

Liu, X., Faes, L., Kale, A. et al. (2019) ‘A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis’, *The Lancet Digital Health*, 1, pp. e271–e297. Available at: [https://doi.org/10.1016/S2589-7500\(19\)30123-2](https://doi.org/10.1016/S2589-7500(19)30123-2)

Lin, T.Y., Goyal, P., Girshick, R., He, K. and Dollár, P., 2017. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision* (pp. 2980-2988). Available at: https://openaccess.thecvf.com/content_iccv_2017/html/Lin_Focal_Loss_for_ICCV_2017_paper.html

Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., Van Der Laak, J.A., Van Ginneken, B. and Sánchez, C.I., 2017. A survey on deep learning in medical image analysis. *Medical image analysis*, 42, pp.60-88. Available at: https://www.sciencedirect.com/science/article/pii/S1361841517301135?casa_token=qzCvsQp8iqIAAAAA:eyAmWPK-6TC_sJTe1JZDi_vpQnXDRXh7ip5b9zS6AUgs0Mn2eNj8BE3jdI44XHfIZw9M7xlT7w

Milletari, F., Navab, N. and Ahmadi, S.A., 2016, October. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In *2016 fourth international conference on 3D vision (3DV)* (pp. 565-571). Ieee. Available at: <https://ieeexplore.ieee.org/abstract/document/7785132>

Minaee, S., Boykov, Y., Porikli, F., Plaza, A., Kehtarnavaz, N. and Terzopoulos, D., 2021. Image segmentation using deep learning: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 44(7), pp.3523-3542. Available at: https://ieeexplore.ieee.org/abstract/document/9356353?casa_token=gkycEMrxXTkAAAAA:wwzVacHrcgridY-zBvSJHjhp0b70sFxBhVmnIJck-LqZ9YJYx-K6kdLVUWMGu16CtEiqDjmg1g

Oktay, O., Schlemper, J., Folgoc, L.L., Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N.Y., Kainz, B. and Glocker, B., 2018. Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*. Available at: <https://arxiv.org/abs/1804.03999>

Painchaud, N., Duchateau, N., Bernard, O. and Jodoin, P.M., 2022. Echocardiography segmentation with enforced temporal consistency. IEEE transactions on medical imaging, 41(10), pp.2867-2878. Available at:

https://ieeexplore.ieee.org/abstract/document/9771186?casa_token=rvVpTUEs0TcAAAAA:tdteltokwzMI26b-vjyFqTG2wILZyzqze2TuLm6E6_J58Z8jsysUy1IXd1HyHLR7fcJcZypTzg

Perez, L. and Wang, J., 2017. The effectiveness of data augmentation in image classification using deep learning. arXiv preprint arXiv:1712.04621. Available at: <https://arxiv.org/abs/1712.04621>

Ronneberger, O., Fischer, P. and Brox, T., 2015, October. U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer-assisted intervention (pp. 234-241). Cham: Springer international publishing. Available at: https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28

Saeed, M., Muhtaseb, R. and Yaqub, M., 2022, July. Contrastive pretraining for echocardiography segmentation with limited data. In Annual conference on medical image understanding and analysis (pp. 680-691). Cham: Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-031-12053-4_50

Shorten, C. and Khoshgoftaar, T.M. (2019) ‘A survey on image data augmentation for deep learning’, Journal of Big Data, 6, article 60. Available at: <https://doi.org/10.1186/s40537-019-0197-0>

Ullah, Z., Hilal, M., Lee, E., Pamucar, D. and Kim, J., 2025. Unified Review and Benchmark of Deep Segmentation Architectures for Cardiac Ultrasound on CAMUS. arXiv preprint arXiv:2601.00839. Available at: <https://arxiv.org/abs/2601.00839>

Van De Vyver, G., Thomas, S., Ben-Yosef, G., Olaisen, S.H., Dalen, H., Løvstakken, L. and Smistad, E., 2024. Toward robust cardiac segmentation using graph convolutional networks. IEEE Access, 12, pp.33876-33888. Available at: <https://ieeexplore.ieee.org/abstract/document/10458930>

Zhou, Z., Rahman Siddiquee, M.M., Tajbakhsh, N. and Liang, J., 2018, September. Unet++: A nested u-net architecture for medical image segmentation. In International workshop on deep learning in medical image analysis (pp. 3-11). Cham: Springer International Publishing. Available at: https://link.springer.com/chapter/10.1007/978-3-030-00889-5_1

Appendix A: Full Experimental Results

The results are shown as segmented figures for easier reading because the experimental results table is large. These figures provide detailed information on segmentation accuracy, boundary precision, and clinical performance metrics.

The results in this appendix reflect the final recorded outputs used in this study. Rerunning the experiments may result in minor differences in the reported metrics because of stochastic elements in the training process, such as data augmentation, sampling techniques, and GPU computation.

Figure A.1: Summary of Model Performance (Dice Scores)

	Experiment_Name	Model	Sampler	Loss_Type	fg_dice_mean
13	AttentionUNet_weighted_focal_dice	AttentionUNet	weighted	focal_dice	0.899058
12	Paper5_UNet_Baseline_weighted_focal_dice	Paper5_UNet_Baseline	weighted	focal_dice	0.897912
28	UNetPP_balanced_focal_dice	UNetPP	balanced	focal_dice	0.892549
26	Paper5_UNet_Baseline_balanced_focal_dice	Paper5_UNet_Baseline	balanced	focal_dice	0.892523
32	Recommended_DeepLabV3_improved_ce_dice	Recommended_DeepLabV3	improved	ce_dice	0.891949
9	Paper2_DeepLabV3_weighted_focal_dice	Paper2_DeepLabV3	weighted	focal_dice	0.888602
1	Paper1_UNet_weighted_non_focal	Paper1_UNet	weighted	non_focal	0.887068
24	Paper3_UNet_CE+Dice_balanced_focal_dice	Paper3_UNet_CE+Dice	balanced	focal_dice	0.886505
23	Paper2_DeepLabV3_balanced_focal_dice	Paper2_DeepLabV3	balanced	focal_dice	0.885914
0	debug_test_improved_sampler	Unknown	improved	unknown	0.885849
22	Paper1_UNet_balanced_focal_dice	Paper1_UNet	balanced	focal_dice	0.885717
21	UNetPP_balanced_non_focal	UNetPP	balanced	non_focal	0.882931
3	Paper3_UNet_CE+Dice_weighted_non_focal	Paper3_UNet_CE+Dice	weighted	non_focal	0.882507
31	Recommended_UNetPP_improved_ce_dice	Recommended_UNet	improved	ce_dice	0.882483
10	Paper3_UNet_CE+Dice_weighted_focal_dice	Paper3_UNet_CE+Dice	weighted	focal_dice	0.881434
17	Paper3_UNet_CE+Dice_balanced_non_focal	Paper3_UNet_CE+Dice	balanced	non_focal	0.881430
7	UNetPP_weighted_non_focal	UNetPP	weighted	non_focal	0.881425
8	Paper1_UNet_weighted_focal_dice	Paper1_UNet	weighted	focal_dice	0.881311
27	AttentionUNet_balanced_focal_dice	AttentionUNet	balanced	focal_dice	0.881306
14	UNetPP_weighted_focal_dice	UNetPP	weighted	focal_dice	0.880547
2	Paper2_DeepLabV3_weighted_non_focal	Paper2_DeepLabV3	weighted	non_focal	0.878406
15	Paper1_UNet_balanced_non_focal	Paper1_UNet	balanced	non_focal	0.876962
30	Recommended_AttentionUNet_improved_ce_dice	Recommended_AttentionUNet	improved	ce_dice	0.875640
11	Paper4_ResUNet_weighted_focal_dice	Paper4_ResUNet	weighted	focal_dice	0.874872
6	AttentionUNet_weighted_non_focal	AttentionUNet	weighted	non_focal	0.874703
33	Recommended_ResUNet_improved_ce_dice	Recommended_ResUNet	improved	ce_dice	0.874423
20	AttentionUNet_balanced_non_focal	AttentionUNet	balanced	non_focal	0.873554
25	Paper4_ResUNet_balanced_focal_dice	Paper4_ResUNet	balanced	focal_dice	0.872836
16	Paper2_DeepLabV3_balanced_non_focal	Paper2_DeepLabV3	balanced	non_focal	0.871993
29	Recommended_UNet_improved_ce_dice	Recommended_UNet	improved	ce_dice	0.870465
4	Paper4_ResUNet_weighted_non_focal	Paper4_ResUNet	weighted	non_focal	0.866946
18	Paper4_ResUNet_balanced_non_focal	Paper4_ResUNet	balanced	non_focal	0.865390
19	Paper5_UNet_Baseline_balanced_non_focal	Paper5_UNet_Baseline	balanced	non_focal	0.859965
5	Paper5_UNet_Baseline_weighted_non_focal	Paper5_UNet_Baseline	weighted	non_focal	0.850997

Figure A.2: Class-wise Dice Scores (LV, MYO, LA)

dice_LV	dice_MYO	dice_LA
0.931915	0.868438	0.896820
0.931240	0.868719	0.893777
0.931267	0.866815	0.879565
0.929988	0.865292	0.882289
0.922532	0.859167	0.894148
0.926684	0.855957	0.883166
0.932291	0.855179	0.873735
0.921746	0.858094	0.879675
0.925354	0.854544	0.877842
0.926350	0.854688	0.876509
0.920768	0.855879	0.880506
0.930190	0.839823	0.878781
0.927066	0.834918	0.885537
0.918005	0.848297	0.881147
0.924796	0.844442	0.875062
0.924650	0.849803	0.869836
0.927332	0.853994	0.862948
0.918490	0.843817	0.881627
0.923954	0.850487	0.869477
0.924239	0.846501	0.870901
0.922054	0.855483	0.857681
0.913285	0.845285	0.872316
0.925586	0.838102	0.863233
0.912363	0.844731	0.867522
0.921968	0.837274	0.864866
0.915550	0.843975	0.863743
0.918992	0.840376	0.861295
0.915211	0.841114	0.862181
0.925567	0.849643	0.840770
0.914017	0.835025	0.862353
0.922526	0.834847	0.843466
0.911320	0.817250	0.867601
0.920854	0.810715	0.848326
0.910660	0.797117	0.845215

Figure A.3: Boundary Metrics (HD95)

hd95_LV	hd95_MYO	hd95_LA
12.399421	13.421245	14.901704
12.505347	14.016629	14.715942
12.489592	13.602149	17.327685
12.870610	14.440309	16.224300
13.159342	14.934508	14.813184
13.220529	15.415802	17.712814
12.068452	15.082281	17.917675
13.434886	15.354987	15.853364
13.231259	15.215149	16.682473
13.154904	15.433832	16.775996
14.454064	15.571641	17.214205
12.805821	16.223855	18.360970
13.200487	17.208609	16.118475
14.871189	16.263658	16.566822
13.744329	16.581025	15.829486
13.688555	16.403047	16.811498
13.599212	15.975375	20.732716
15.743167	16.172778	16.591225
13.710506	15.550320	17.573803
13.956064	16.865791	17.202876
13.982073	15.295493	19.712145
15.655214	17.138606	18.108849
13.771984	17.161644	17.908952
15.677766	16.313895	16.917126
14.130433	17.778933	18.335588
15.539456	18.262210	20.032180
14.482167	17.910956	19.654035
16.166213	17.510044	18.644726
13.305983	15.372134	23.220467
16.141497	17.888971	20.186497
14.318003	19.261749	21.794709
16.518746	19.435864	18.026298
14.951097	21.316526	21.965512
15.536153	22.399391	22.210817

Figure A.4: Clinical Metrics and Training Performance

EDV_MAE_ml	ESV_MAE_ml	EF_MAE_pct	val_fg_dice_mean	Time_Sec
8.370638	4.194926	3.790712	0.897970	862.645389
8.324658	5.351061	5.529522	0.902098	738.767320
7.269133	4.795529	4.698625	0.897505	1337.717206
8.978902	4.513606	6.595804	0.892181	659.171577
9.569262	6.013656	4.709313	0.894548	607.372726
9.190404	4.785744	5.754926	0.894642	564.519673
7.199464	4.486350	3.943373	0.887445	723.276096
10.348879	6.417120	5.034351	0.891089	400.202659
9.541128	4.726450	4.816204	0.889311	423.932724
8.311235	4.836623	4.560903	0.885733	437.410685
11.392714	6.030836	4.632895	0.887016	513.558274
7.978688	4.501831	4.922111	0.884047	1069.568412
8.308106	5.930189	5.307935	0.879742	253.846524
11.335425	5.552625	5.906104	0.885763	732.647224
7.825630	5.111071	4.904549	0.885384	475.093866
9.152165	5.771957	5.086941	0.884307	365.420632
9.937352	4.741717	4.249072	0.883333	863.770573
8.327727	5.397185	5.302430	0.883510	367.554628
10.141085	5.467118	6.490335	0.882877	401.668192
8.539922	5.613899	6.124768	0.883558	534.647972
11.708749	6.587299	6.292416	0.881583	553.828789
12.583537	7.477686	5.141313	0.879458	400.794926
8.346056	6.005382	5.832404	0.879243	453.622536
7.854141	6.089698	6.959167	0.879161	471.245902
8.843156	4.642683	5.963662	0.876439	402.280361
10.884519	6.633164	7.714696	0.881816	612.041224
11.430676	6.215106	4.249027	0.875162	554.094469
9.330591	7.407170	6.670961	0.874946	374.717454
9.684546	4.494638	4.361023	0.873621	464.017145
10.106978	7.613558	5.991773	0.872965	291.211622
8.231262	4.482990	5.844187	0.869154	651.946446
9.992455	5.494260	4.867926	0.863644	515.431461
8.036224	5.174936	4.516816	0.860726	327.561598
9.481331	6.452263	5.693719	0.851885	252.938201